F78AB Actuarial and Financial Mathematics B - Sample solutions to May 2015 Exam

1. (a)

$$PV(0) = 100 \exp\left\{-\int_0^5 0.005s ds\right\} = 100e^{-0.0625} = £93.94.$$

(b) With $\rho(t) := 10t$ paid from $t = 3$ to $t = 6$ only, the accumulated value at time 6 is

$$AV(0) = \int_3^6 \rho(t) e^{\int_t^6 \delta(s) ds} dt = \int_3^6 10te^{\int_t^6 0.005s ds} dt.$$

As $\frac{d}{dt}\int_t^6 0.005s ds = -0.005t$,

$$AV(0) = \frac{10}{-0.005}\int_3^6 (-0.005t)e^{\int_t^6 0.005s ds} dt = -2000\int_3^6 \frac{d}{dt}\left(e^{\int_t^6 0.005s ds}\right) dt.$$

Hence

$$AV(0) = -2000\left[e^{\int_t^6 0.005s ds}\right]_{t=3}^{t=6} = -2000\left[1 - e^{0.0675}\right] = 139.66.$$

2. (a) Short-selling a financial asset means selling it today with the intention of buying it later.

(b) Denote by K the forward price and by S_t the price of the stock at time t. The payoff on one long forward is $S_{7/12} - K$.

To replicate the payoff, at time 0 borrow £60 and buy one unit of the stock for £60. The net cost is £0.

At time $t = 3/12$, deposit the £2 dividend in the bank account. Similarly, at time $t = 6/12$, deposit the £2 dividend in the bank account.

Then the amount in the bank account at $t = 7/12$ is

$$-60e^{0.04\times 7/12} + 2e^{0.04\times(7-3)/12} + 2e^{0.04\times(7-6)/12} = -61.416 + 2.027 + 2.007 = -57.382$$

and the value of the stock is $S_{7/12}$. Setting $K = 57.382$, this is a replicating portfolio for the long forward.

This assumes there is no arbitrage.

3. (a) (i) Convert the money cashflows to real cashflows:

Time t (years)	0	1	2	3
Retail price index	341.4	366.6	382.0	401.8
Money cashflow (£000s)	-54	20	20	20
Real cashflow (£000s) at time 0 real values	$-54 \times \frac{341.4}{341.4} = -54$	$20 \times \frac{341.4}{366.6}$ 18.625	$20 \times \frac{341.4}{382.0}$ 17.874	$20 \times \frac{341.4}{401.8}$ 16.994

Denote the real yield by j. Then find the positive root of $f(j) := -54(1+j)^3 + 18.625(1+j)^2 + 17.874(1+j) + 16.994$ by trial-and-error.
A first estimate is

$$-54(1+3j)+18.625(1+2j)+17.874(1+j)+16.994 \approx 0 \quad \Rightarrow j \approx \frac{0.507}{106.876} = -0.5\%.$$

Then $f(-0.5\%) = 0.024$ and $f(0\%) = -0.51$. Hence the real yield is -0.5% p.a. to the nearest 0.5%.

(ii) Denote the money yield by i. Then find the positive root of $g(i) := -54(1+i)^3 + 20(1+i)^2 + 20(1+i) + 20$ by trial-and-error.
A first estimate is

$$-54(1+3i)+20(1+2i)+20(1+i)+20 \approx 0 \quad \Rightarrow i \approx \frac{6}{82} = 5.9\%.$$

Then $g(6\%) = -0.64$, $g(5.5\%) = -0.05$, $g(5\%) = 0.54$. Hence the money yield is 5.5% p.a. to the nearest 0.5%.

(b) Disagree. If inflation is constant, then the annual inflation $\bar{q}$ satisfies $1 + \bar{q} = \frac{1+i}{1+j}$, where i is the money yield and j is the real yield. However, this is no longer true if inflation is not constant.

The difference in yields is approximately 6% p.a. Average inflation over the three years is $(\frac{401.8}{341.4})^{1/3} - 1 = 5.6\%$ p.a. The discrepancy between these two values suggests that inflation is not constant over the three years. Calculating the annual inflation over the three years, we see that this is indeed the case:

Time t (years)	0	1	2	3
Retail price index	341.4	366.6	382.0	401.8
Annual inflation from $t-1$ to t		7.4%	4.2%	5.2%

4. (a) As $\mathbb{E}\left(i_1^2\right) = \text{var}(i_1) + (\mathbb{E}(i_1))^2 = (0.15)^2 + (0.07)^2 = 0.0274$, and from the given moments formula,

$$e^{\mu+\frac{1}{2}\sigma^2} = \mathbb{E}\,(1+i_1) = 1.07 \qquad \text{and} \qquad e^{2\mu+2\sigma^2} = \mathbb{E}\,(1+i_1)^2 = 1.1674.$$

Hence

$$e^{\sigma^2} = \frac{1.1674}{e^{2\mu+\sigma^2}} = \frac{1.1674}{1.07^2} \quad \Rightarrow \quad \sigma = 0.139505,$$

and $\mu = \ln(1.07) - \frac{1}{2}(0.139505)^2 = 0.057928$.

(b) As $5\,000S_{20} = 5\,000\prod_{n=1}^{20}(1+i_n)$, then $\ln(5\,000S_{20}) = \ln(5\,000) + \sum_{n=1}^{20}\ln(1+i_n)$.
Then as $\ln(1+i_n) \sim \mathcal{N}(\mu, \sigma^2)$ for $n = 1, 2, \ldots, 20$, and the sum of independent normally distributed random variables is itself normally distributed with mean the sum of the means, and variance the sum of the variances,

$$\ln(5\,000S_{20}) \sim \mathcal{N}(\ln(5\,000)+20\mu, 20\sigma^2) = \mathcal{N}(9.675753, (0.623885)^2).$$

Thus $5\,000S_{20}$ is lognormally-distributed with parameters 9.675753 and 0.623885.

(c) i. Denote by $\bar{x}$ the mean accumulated amount. Then $\bar{x}$ satisfies

$$\bar{x} = \mathbb{E}\,(5\,000S_{20}) = e^{9.675753+\frac{1}{2}(0.623885)^2} = 19\,348.48.$$

ii. Denote by $\bar{\delta}$ the mean annual force of interest. Then

$$e^{20\bar{\delta}} = \frac{19\,348.48}{5\,000} \quad \Rightarrow \quad \bar{\delta} = 0.067659.$$

iii. Denote by x_m the median accumulated amount. Then x_m satisfies

$$0.5 = \mathbb{P}[5\,000S_{20} \le x_m] = \Phi\left(\frac{\ln(x_m) - 9.675753}{0.623885}\right).$$

As $\Phi(0) = 0.5$, then $x_m = e^{9.675753} = 15\,926.71$.

iv. Denote by δ_m the median annual force of interest. Then

$$e^{20\delta_m} = \frac{15\,926.71}{5\,000} \quad \Rightarrow \quad \delta_m = 0.057928.$$

(d) Any reasonable answer gains credit. There should be a reference to investment uncertainty (risk) in the answer.
e.g. At the end of the 20 years, there is a 50% chance of the annualised return being higher than 5.7928% p.a, and a 50% chance of being lower.

5. (a) With $v = \frac{1}{1.05}$, the duration equals $\frac{\sum_{k=1}^{12} kv^k}{\sum_{k=1}^{12} v^k} = \frac{(Ia)_{\overline{12}|}}{a_{\overline{12}|}}$. As

$$a_{\overline{12}|} = \frac{1-v^{12}}{i} = 8.86325,\ \ddot{a}_{\overline{12}|} = \frac{1-v^{12}}{d} = 9.30641,\ (Ia)_{\overline{12}|} = \frac{\ddot{a}_{\overline{12}|} - 12v^{12}}{i} = 52.48722,$$

the duration is $\frac{52.487}{8.863} = 5.922$ years.

(b) Work in £million. Present value of assets PV_A and liabilities PV_L:

$$\begin{aligned}\mathrm{PV}_A =& a_{\overline{12}|} + Xv^N = 8.863 + Xv^N.\\ \mathrm{PV}_L =& 10v^7 + 11v^9 = 14.198.\end{aligned}$$

Duration of assets τ_A and liabilities τ_L:

$$\begin{aligned}\tau_A =& \frac{(Ia)_{\overline{12}|} + N \cdot Xv^N}{\mathrm{PV}_A} = \frac{52.487 + N \cdot Xv^N}{\mathrm{PV}_A}.\\ \tau_L =& \frac{7 \cdot 10v^7 + 9 \cdot 11v^9}{\mathrm{PV}_L} = \frac{113.564}{\mathrm{PV}_L}.\end{aligned}$$

Convexity of assets conv_A and liabilities conv_L:

$$\begin{aligned}\mathrm{conv}_A =& \frac{\sum_{k=1}^{12} k^2 v^k + N^2 \cdot Xv^N}{\mathrm{PV}_A} = \frac{414.647 + N^2 \cdot Xv^N}{\mathrm{PV}_A}.\\ \mathrm{conv}_L =& \frac{7^2 \cdot 10v^7 + 9^2 \cdot 11v^9}{\mathrm{PV}_L} = \frac{922.580}{\mathrm{PV}_L}.\end{aligned}$$

Equate present values:

$$8.863 + Xv^N = 14.198 \quad \Rightarrow \quad Xv^N = 5.335.$$

Equate durations to solve for N:

$$52.487 + N \cdot Xv^N = 113.564 \quad \Rightarrow \quad N = \frac{113.564 - 52.487}{5.335} = 11.448.$$

Solve for X:

$$Xv^N = 5.335 \quad \Rightarrow \quad X = 5.335(1.05)^{11.448} = 9.326.$$

Check convexity condition for immunisation to hold, i.e. if $\mathrm{conv}_A \geq \mathrm{conv}_L$.

$$\mathrm{conv}_A \cdot \mathrm{PV}_A = 414.647 + (11.448)^2 \cdot (5.335) = 1\,113.935 > 922.580 = \mathrm{conv}_L \cdot \mathrm{PV}_L.$$

In summary, by buying £9.326 million nominal of an 11.448-year zero-coupon bond, the company is immunised against small changes in the interest rate.

6. (a) Assuming no arbitrage holds , and as

$$y_2 = 0.006 + \frac{12}{1000} = 0.018 \quad \text{and} \quad y_7 = 0.006 + \frac{42}{1000} = 0.048,$$

then

$$f_{2,7} = \left(\frac{(1+y_7)^7}{(1+y_2)^2}\right)^{1/5} - 1 = \left(\frac{1.048^7}{1.018^2}\right)^{1/5} - 1 = 0.060246.$$

(b)

Time n	0	1	2	3	4
Cashflow C_n (£000s)	-65	5	10	30	30
y_n	0	0.012	0.018	0.024	0.030
Discount factor	1	$\frac{1}{1.012}$	$\frac{1}{1.018^2}$	$\frac{1}{1.024^3}$	$\frac{1}{1.030^4}$
Discounted cashflow	-65	4.941	9.649	27.940	26.655
Cum. disc. cashflow	-65	-60.059	-50.410	-22.470	4.185

The net present value is £4 185, the sum of the discounted cashflows.

(c) From the calculations in the table above, the discounted payback period is 4 years since this the first integer time at which the net present value becomes positive.